MA416

Lecture 05 – September 16, 2025
Distribution Testing

Dr. Matthew N. Moore

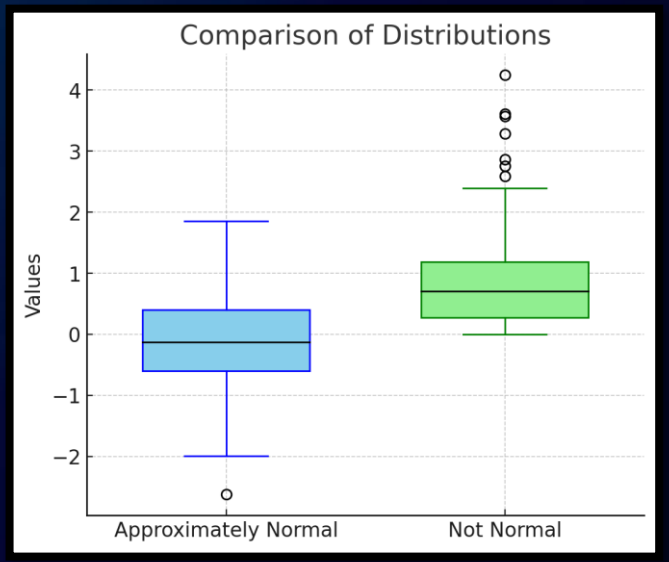
Summary of Tests So Far…

1. Test on $H_0: \mu = \mu_0$ with T_{stat}
   - Requires normality of the population (and …)

2. Test on $H_0: \mu_1 = \mu_2$ with T_{stat}
   - Requires normality of the populations (and …)

3. Test on $H_0: \sigma_1^2 = \sigma_2^2$ with F_{stat}
   - Requires normality of the populations (and …)

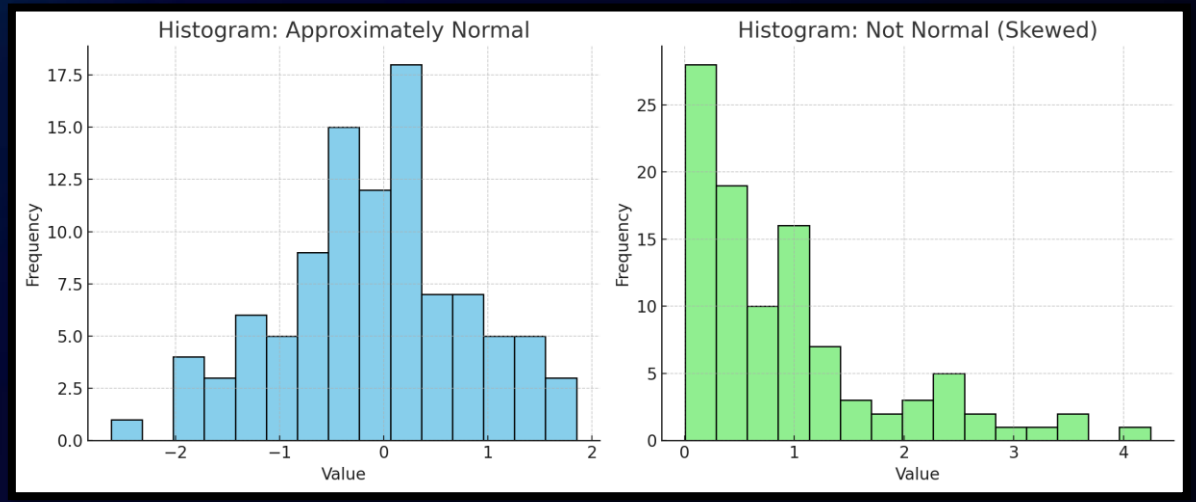
4. Testing any hypothesis with user-generated statistics
   - Doesn't require normality, but statistical power is at risk.



Q) How to assess normality?

Qualitative methods include:

- Histograms
- Boxplots

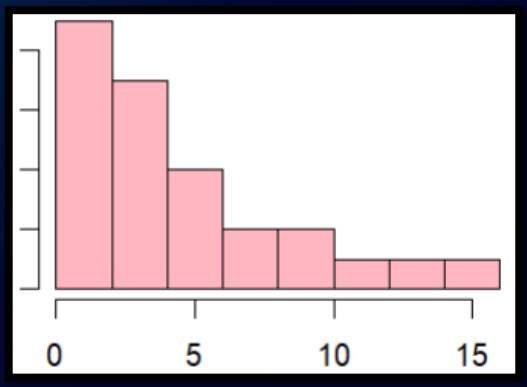


Histogram (or Boxplot) Comparisons

Example: Someone claims that the population is exponentially distributed with mean 4.7.

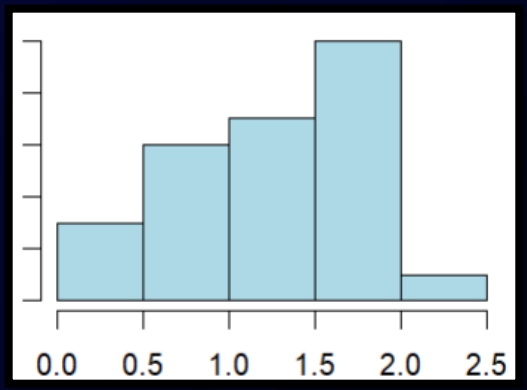
Q) If $n = 27$, what is the ideal distribution of these points to not reject this claim?

```
n = 27
Fx = seq(1, n, 1)/(n+1)
S_ideal = qexp(Fx, 1/4.7, lower.tail=T)
hist(S, col='lightpink')
```



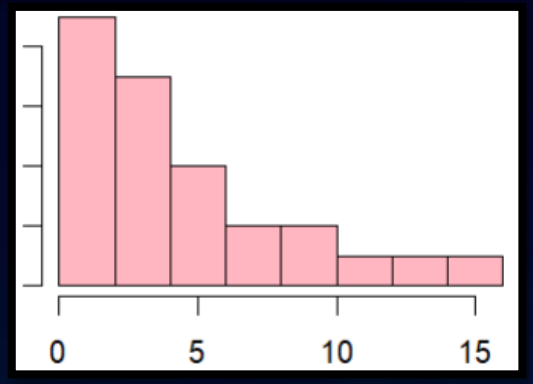
Q2) Do we have evidence to believe our sample is associated to this distribution?

```
S_actual = mydata$Prior_Experience[1:27]
hist(S_actual, col='lightblue')
```

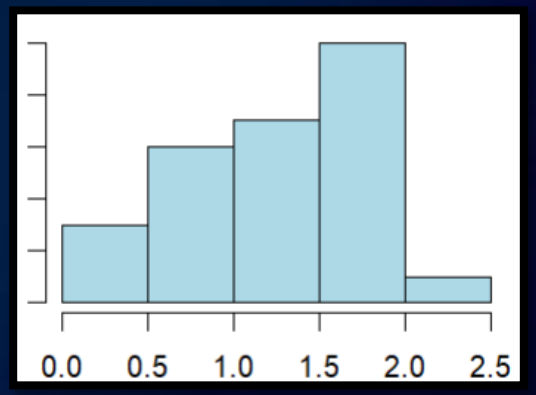


Quantile-Quantile (QQ) Plots

```
n = 27
Fx = seq(1, n, 1)/(n+1)
S_ideal = qexp(Fx, 1/4.7, lower.tail=T)
hist(S, col='lightpink')
```




```
S_actual = mydata$Prior_Experience[1:27]
hist(S_actual, col='lightblue')
```

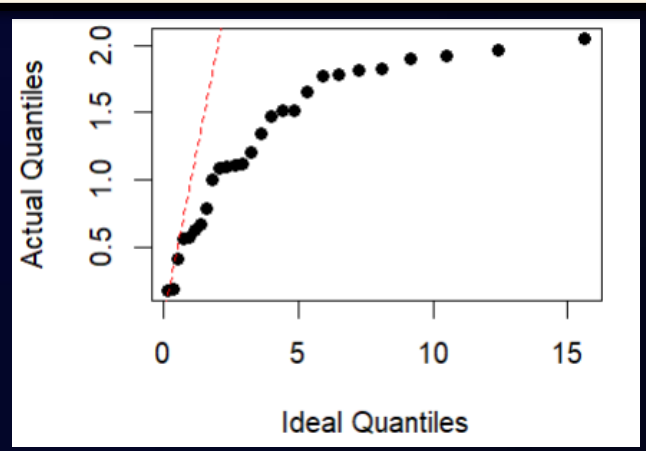


A "quantile-quantile (QQ) plot" is a qualitative method that directly compares the quantiles of a data set to the quantiles of the hypothesized (null) distribution.

Decision Rule (for $H_0: P = P_0$)

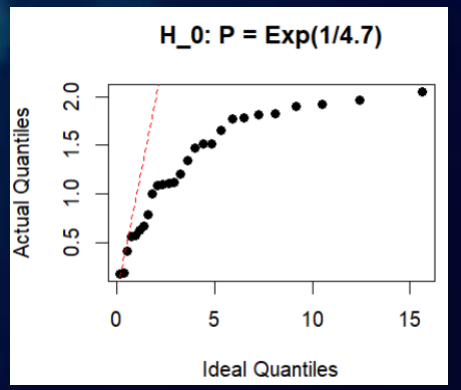
- If the quantiles are close, fail to reject H_0
- If the quantiles are far, reject H_0

```
plot(S_ideal, sort(S_actual), pch=16, col='black',
     xlab='Ideal Quantiles', ylab='Actual Quantiles')
abline(0,1, lty=2, col='red')
```



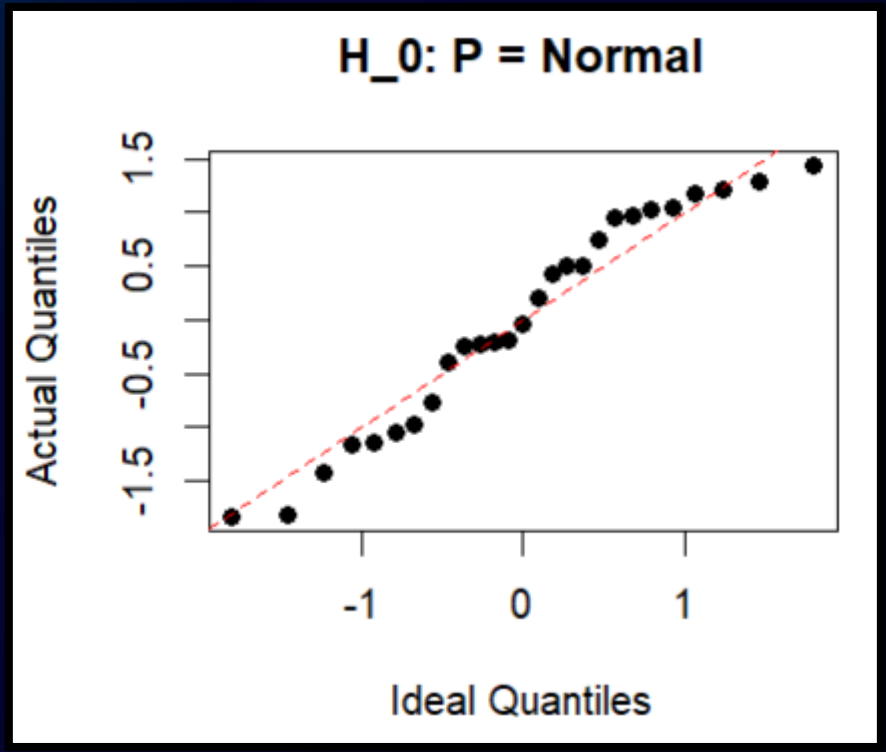
Quantile-Quantile (QQ) Plots

Example: Someone claims that the population is normally distributed (any μ, σ^2 is fine).

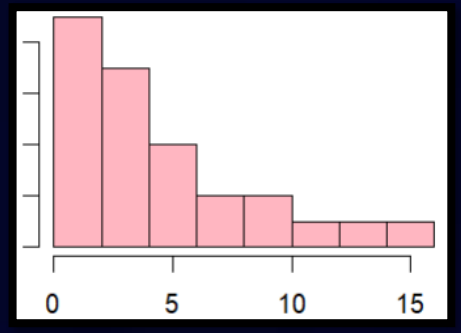


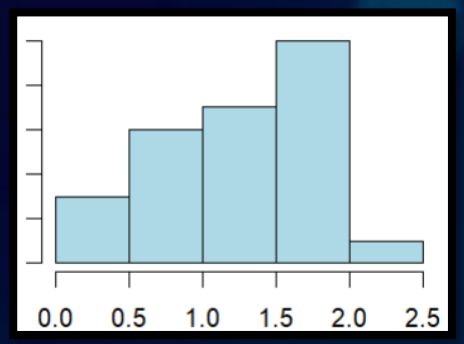

```
X = mydata$Prior_Experience[1:27]
X = sort(X)
S_ideal = qnorm(Fx, 0, 1, lower.tail=T)
S_actual = (X - mean(X))/sd(X)
```

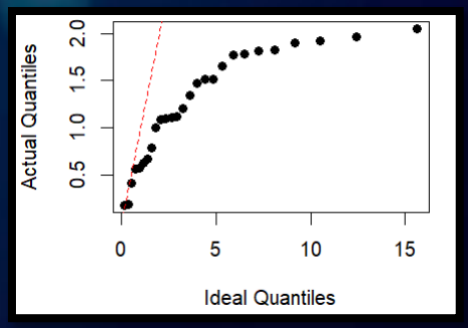
```
plot(S_ideal, sort(S_actual), pch=16, col='black',
     xlab='Ideal Quantiles', ylab='Actual Quantiles',
     main = 'H_0: P = Normal')
abline(0,1, lty=2, col='red')
```



CDF Comparisons







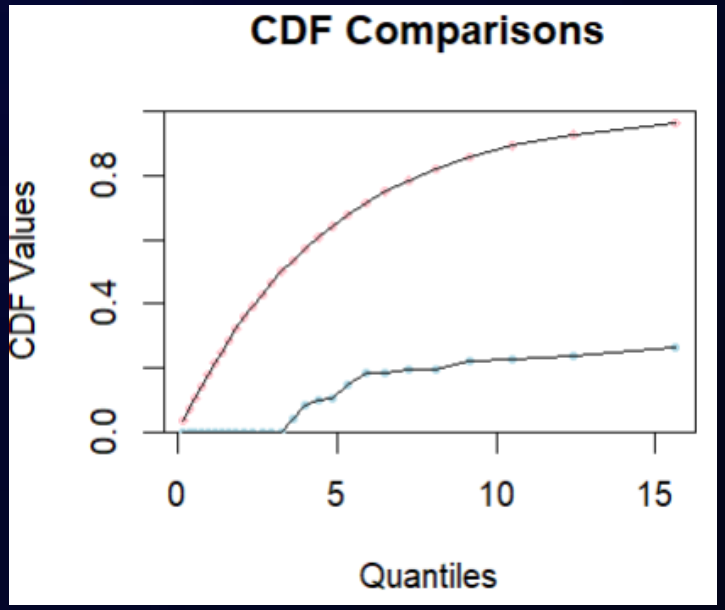
Observation: If the quantiles match one another, then their associated cumulative density function (CDF) values should also match one another.

```
n = 27
Fx = seq(1, n, 1)/(n+1)
S_ideal = qexp(Fx, 1/4.7, lower.tail=T)

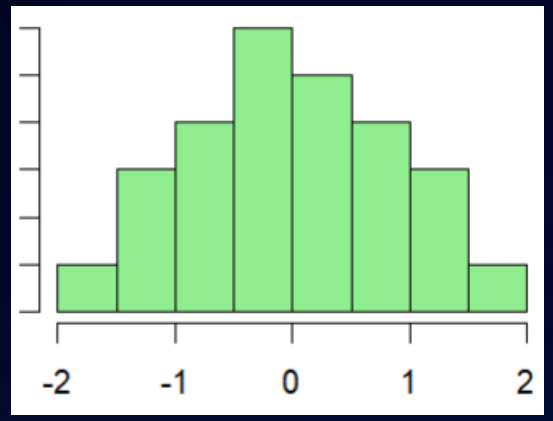
S_actual = sort(mydata$Prior_Experience[1:27])
Psix = pexp(S_actual, 1/4.7, lower.tail=T)

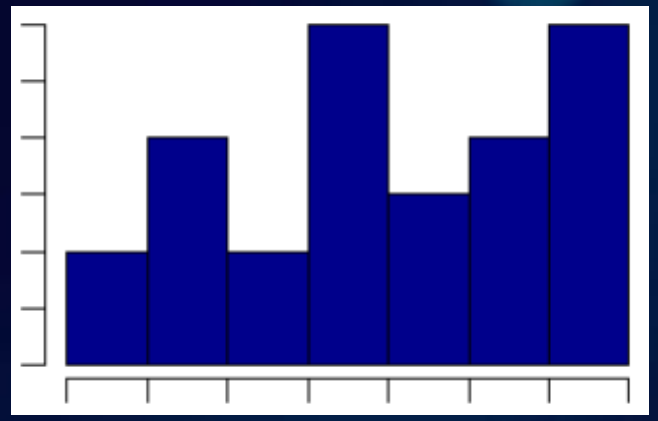
plot(S_ideal, Fx, pch=10, col='lightpink', cex=0.5,
     xlab='Quantiles', ylab='CDF Values', main='CDF Comparisons')
lines(S_ideal, Fx)

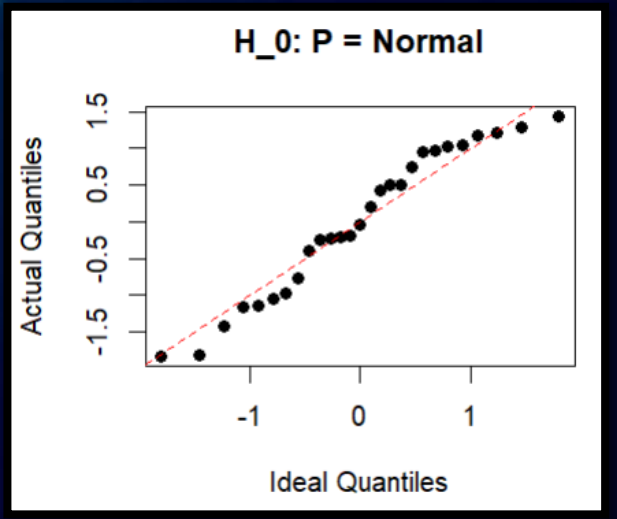
points(S_ideal, Psix, pch=16, col='lightblue', cex=0.5)
lines(S_ideal, Psix)
```



CDF Comparisons

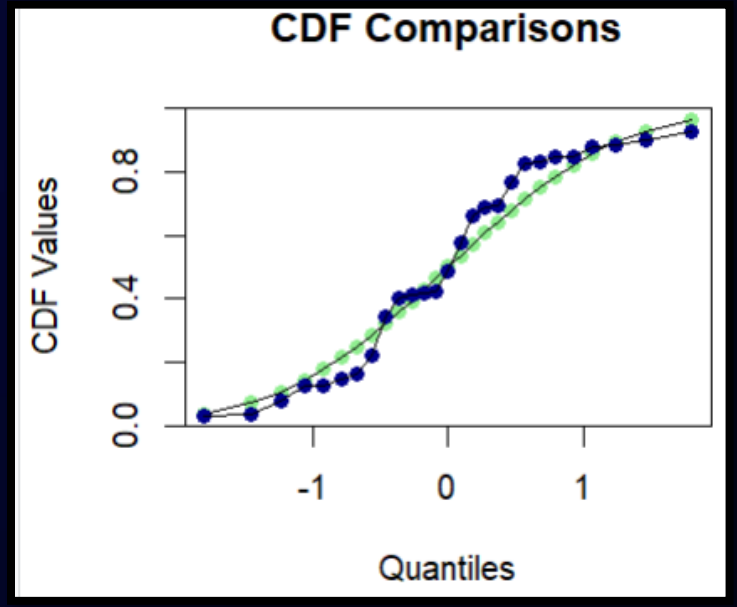






Observation: Boxplots, histograms, quantile-quantile, and CDF comparison plots are all qualitative methods, and hence carry their own subjectivity in terms of interpretation.

Observation 2: If the CDFs are close to one another, the area between them will be close to 0.



Quantitative Metrics for Distribution Testing

Let F_x and ψ_x are the empirical and theoretical CDFs for two different distributions, where $F_x \sim S$ and $\psi_x \sim P_0$ with $S \subseteq P$. Then if:

$$H_0 : P = P_0$$
$$H_a : P \neq P_0$$

and if we define:

$$\star_{stat} := \int_{-\infty}^{\infty} |F_x(t) - \psi_x(t)| \, dt$$

Note that $\left(F_x(t) - \psi_x(t)\right) \in (-1, 1)$ for some $t \in \mathbb{R}$. Therefore:

Decision Rule (for $H_0 : P = P_0$)

- If $\star_{stat} \approx 0$, fail to reject H_0
- If $\star_{stat} \gg 0$, reject H_0

Common Testing Structures for Normality

Definition: A distribution test is called “simple” if both the shape and all of its parameters are specified. A distribution test is called “composite” if the shape is specified, but one or more parameters are not specified.

Special Names Associated to Normal Distributions

Kolmogorov-Smirnov Test

$$H_0: P \sim N(\mu, \sigma^2)$$

Lilliefors’ Test

$$H_0: P \sim N(\text{any } \mu, \text{any } \sigma^2)$$

Common Test Statistics

Option 1: "Kolmogorov-Smirnov" (KS) test statistic

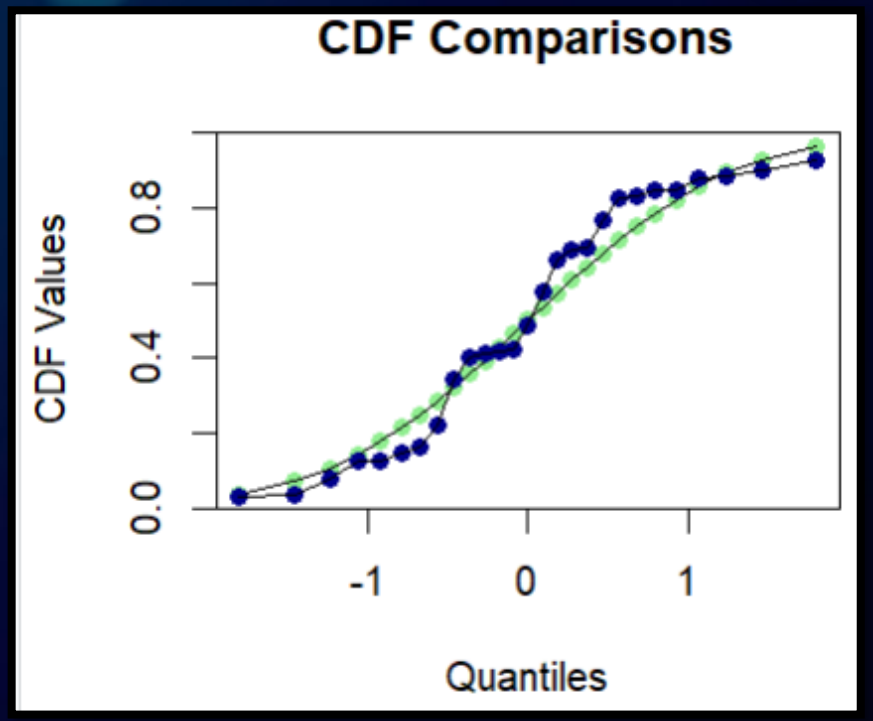
$$KS_{stat} = \max_k |F_k - \psi_k|$$

Option 2: "Cramer-von-Mises" (CvM) test statistic

$$CvM_{stat} = \sum_{k=1}^{n} |F_k - \psi_k|^2$$

Option 3: "Anderson-Darling" (AD) test statistic

$$AD_{stat} = \sum_{k=1}^{n} \frac{|F_k - \psi_k|^2}{F_k \cdot (1 - F_k)}$$



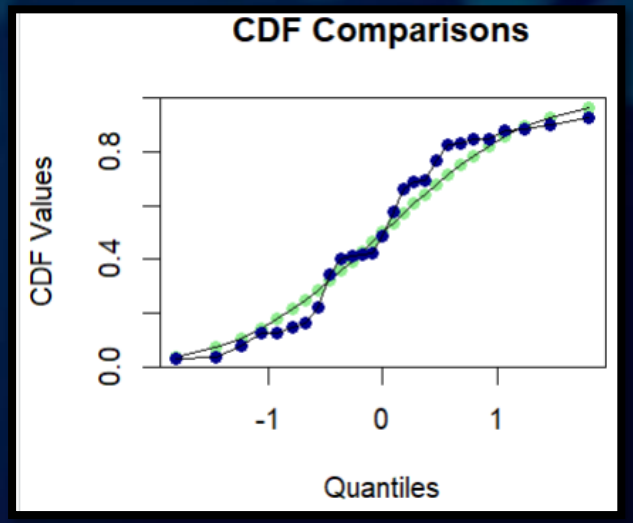
Decision Rule (for $H_0: P = P_0$)

- If $\star_{stat} \approx 0$, fail to reject H_0
- If $\star_{stat} \gg 0$, reject H_0

Example

Claim: Our sample comes from a normally distributed population.

Testing Structure → Lilliefors' (composite) test with a Cramer-von-Mises statistic.

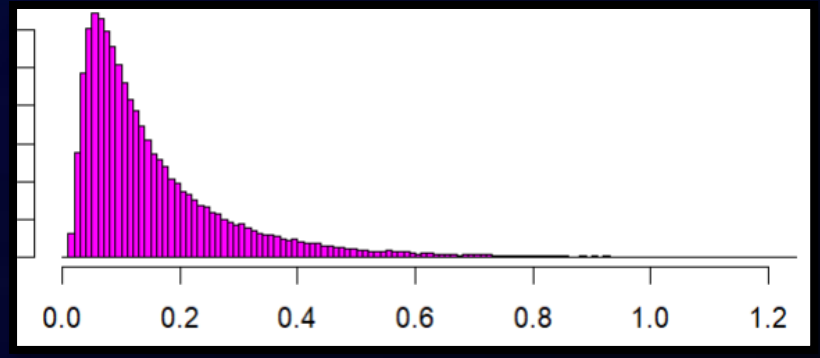



```r
mc_cvm = c()
set.seed(123)
nmc = 100000
for(k in 1:nmc){
  Smc = rnorm(n, 0, 1)
  Smc = sort(Smc)
  Psixmc = pnorm(Smc, 0, 1, lower.tail=T)
  Dkmc = abs(Fx - Psixmc)

  mc_cvm = c(mc_cvm, sum(Dkmc^2))
} # Run Time ~ 15 seconds


hist(mc_cvm, breaks=200, xlim=c(0, 1.8),
     col='magenta', main='Empirical MC CvM Distribution')
max(mc_cvm) # ~= 1.8053
```

```r
Dk = abs(Fx - Psix)
CvM_stat = sum(Dk^2) # ~= 0.0810
```




```r
alpha = 0.05

emp_cvm_crit = quantile(mc_cvm, 1-0.05) # ~= 0.4415
emp_pval = mean(mc_cvm >= CvM_stat) # ~= 0.6641
```